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HACK THE BOX SHERLOCK - SOC NOXIOUS

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Noxious]
└─$ ls
capture.pcap  noxious.zip
```

For this Sherlock, there is only 1 file under the zip file, which is a PCAP file called capture.pcap. Here is what it looks like after opening it in Wireshark.

Apply a display filter ... <Ctrl-/> + DNS HTTP TCP SYN GETPOST HTTP SUS USER AGENT SUS FILES CMD/PWSHELL MZ FTP REQ TLS CLIENT HELL

No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
1	0.0000…	172.17.79.130	172.17.79.1		SSH	178	Server: Encrypted packet (len=124)	0	
2	0.0078…	172.17.79.1	172.17.79.130		TCP	60	60381 → 22 [ACK] Seq=1 Ack=125 Win=4098 Len=0	0	
3	0.0834…	172.17.79.1	239.255.255.250		SSDP	179	M-SEARCH * HTTP/1.1		239.255.255.250:1900
4	0.9514…	172.17.79.136	34.160.236.64		TCP	60	51320 → 443 [ACK] Seq=1 Ack=1 Win=63336 Len=1	1	
5	0.9514…	34.160.236.64	172.17.79.136		TCP	60	443 → 51320 [ACK] Seq=1 Ack=2 Win=64240 Len=0	1	
6	1.2429…	172.17.79.136	216.58.204.142		UDP	1288	50231 → 443 Len=1246		
7	1.2432…	172.17.79.136	216.58.204.142		UDP	771	50231 → 443 Len=729		
8	1.2862…	172.17.79.1	224.0.0.22		IGMPv3	60	Membership Report / Leave group 224.168.100.1		
9	1.3113…	172.17.79.136	35.208.249.213		TCP	60	51308 → 443 [ACK] Seq=1 Ack=1 Win=62852 Len=1	2	
10	1.3113…	35.208.249.213	172.17.79.136		TCP	60	443 → 51308 [ACK] Seq=1 Ack=2 Win=64240 Len=0	2	
11	1.3836…	216.58.204.142	172.17.79.136		UDP	69	443 → 50231 Len=27		
12	1.3940…	172.17.79.1	224.0.0.22		IGMPv3	60	Membership Report / Join group 224.168.100.1 for any sources		
13	1.4020…	216.58.204.142	172.17.79.136		UDP	490	443 → 50231 Len=448		
14	1.4020…	216.58.204.142	172.17.79.136		UDP	146	443 → 50231 Len=104		
15	1.4026…	172.17.79.136	216.58.204.142		UDP	77	50231 → 443 Len=35		
16	1.4287…	172.17.79.136	216.58.204.142		UDP	75	50231 → 443 Len=33		
17	1.5713…	216.58.204.142	172.17.79.136		UDP	66	443 → 50231 Len=24		
18	1.5869…	172.17.79.136	104.18.20.126		TLSv1.2	182	Application Data	3	
19	1.5869…	172.17.79.136	104.18.20.126		TLSv1.2	93	Application Data	3	
20	1.5870…	104.18.20.126	172.17.79.136		TCP	60	443 → 51831 [ACK] Seq=1 Ack=129 Win=64240 Len=0	3	
21	1.5870…	104.18.20.126	172.17.79.136		TCP	60	443 → 51831 [ACK] Seq=1 Ack=168 Win=64240 Len=0	3	
22	1.6389…	96.46.186.186	172.17.79.136		TLSv1.2	93	Application Data	4	
23	1.6398…	172.17.79.136	96.46.186.186		TCP	60	51607 → 443 [FIN, ACK] Seq=1 Ack=40 Win=63177 Len=0	4	
24	1.6398…	96.46.186.186	172.17.79.136		TCP	60	443 → 51607 [ACK] Seq=40 Ack=2 Win=64239 Len=0	4	
25	1.6401…	96.46.186.186	172.17.79.136		TLSv1.2	78	Application Data	4	
26	1.6402…	172.17.79.136	96.46.186.186		TCP	60	51607 → 443 [RST, ACK] Seq=2 Ack=64 Win=0 Len=0	4	
27	1.7318…	2.17.140.234	172.17.79.136		TLSv1.2	78	Application Data	5	
28	1.7328…	172.17.79.136	2.17.140.234		TCP	60	51603 → 443 [FIN, ACK] Seq=1 Ack=25 Win=64216 Len=0	5	
29	1.7328…	2.17.140.234	172.17.79.136		TCP	60	443 → 51603 [ACK] Seq=25 Ack=2 Win=64239 Len=0	5	
30	1.7328…	2.17.140.234	172.17.79.136		TCP	60	443 → 51603 [FIN, PSH, ACK] Seq=25 Ack=2 Win=64239 Len=0	5	
31	1.7331…	172.17.79.136	2.17.140.234		TCP	60	51603 → 443 [ACK] Seq=2 Ack=26 Win=64216 Len=0	5	
32	1.7448…	172.17.79.1	224.0.0.22		IGMPv3	60	Membership Report / Join group 224.168.100.1 for any sources		
33	1.7717…	104.18.20.126	172.17.79.136		TLSv1.2	194	Application Data	3	
34	1.7717…	104.18.20.126	172.17.79.136		TLSv1.2	661	Application Data	3	

1. Its suspected by the security team that there was a rogue device in Forela's internal network running responder tool to perform an LLMNR Poisoning attack. Please find the malicious IP Address of the machine.

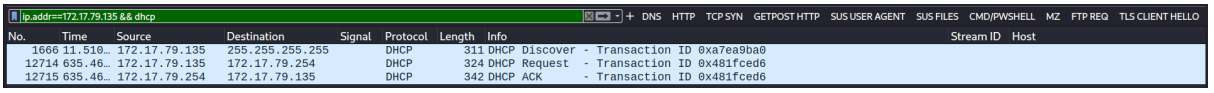
   After googling what is LLMNR Poisoning Attack, An LLMNR (Link-Local Multicast Name Resolution) poisoning attack is a man-in-the-middle (MitM) technique used by threat actors to steal user credentials, such as password hashes, on a local network

udp.port == 5355 + DNS HTTP TCP SYN GETPOST HTTP SUS USER AGENT SUS FILES CMD/PWSHELL MZ FTP REQ TLS CLIENT HELLO

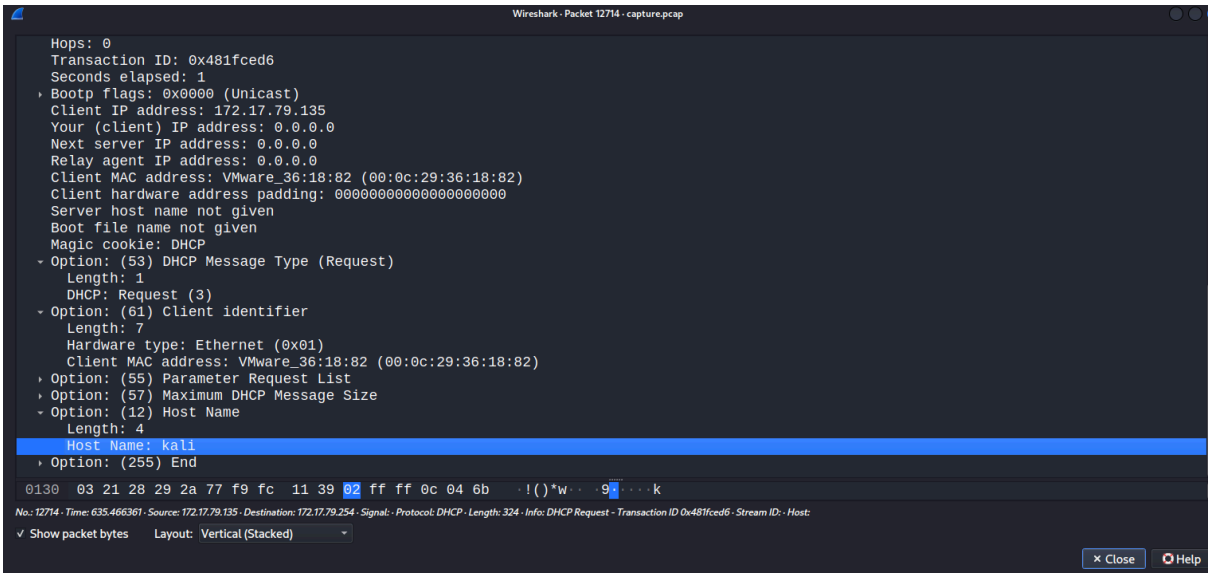
No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
9262	68.433…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0xe708 A DCC01		
9263	68.433…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0xe708 A DCC01		
9264	68.433…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0x2c01 AAAA DCC01		
9265	68.433…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0x2c01 AAAA DCC01		
9268	68.438…	fe80::2068:fe8…	fe80::7994:1860…		LLMNR	106	Standard query response 0xe708 A DCC01 A 172.17.79.135		
9269	68.440…	172.17.79.135	172.17.79.136		LLMNR	86	Standard query response 0xe708 A DCC01 A 172.17.79.135		
9274	68.441…	fe80::2068:fe8…	fe80::7994:1860…		LLMNR	118	Standard query response 0x2c01 AAAA DCC01 AAAA fe80::2068:fe84:5fc…		
9277	68.445…	172.17.79.135	172.17.79.136		LLMNR	98	Standard query response 0x2c01 AAAA DCC01 AAAA fe80::2068:fe84:5fc…		
9301	68.470…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0x3ca4 A DCC01		
9302	68.470…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0x3ca4 A DCC01		
9303	68.470…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0x2dd6 AAAA DCC01		
9304	68.471…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0x2dd6 AAAA DCC01		
9305	68.473…	fe80::2068:fe8…	fe80::7994:1860…		LLMNR	106	Standard query response 0x3ca4 A DCC01 A 172.17.79.135		
9307	68.476…	172.17.79.135	172.17.79.136		LLMNR	86	Standard query response 0x3ca4 A DCC01 A 172.17.79.135		
9309	68.479…	fe80::2068:fe8…	fe80::7994:1860…		LLMNR	118	Standard query response 0x2dd6 AAAA DCC01 AAAA fe80::2068:fe84:5fc…		
9315	68.483…	172.17.79.135	172.17.79.136		LLMNR	98	Standard query response 0x2dd6 AAAA DCC01 AAAA fe80::2068:fe84:5fc…		
9328	68.508…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0x9970 A DCC01		
9329	68.508…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0x9970 A DCC01		
9331	68.508…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0xc75b AAAA DCC01		
9332	68.508…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0xc75b AAAA DCC01		
9333	68.509…	fe80::2068:fe8…	fe80::7994:1860…		LLMNR	106	Standard query response 0x9970 A DCC01 A 172.17.79.135		
9337	68.512…	172.17.79.135	172.17.79.136		LLMNR	86	Standard query response 0x9970 A DCC01 A 172.17.79.135		
9345	68.519…	fe80::2068:fe8…	fe80::7994:1860…		LLMNR	118	Standard query response 0xc75b AAAA DCC01 AAAA fe80::2068:fe84:5fc…		
9346	68.524…	172.17.79.135	172.17.79.136		LLMNR	98	Standard query response 0xc75b AAAA DCC01 AAAA fe80::2068:fe84:5fc…		
9489	78.301…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0x89b8 A DCC01		
9490	78.301…	172.17.79.136	224.0.0.252		LLMNR	65	Standard query 0x89b8 A DCC01		
9491	78.301…	fe80::7994:186…	ff02::1:3		LLMNR	85	Standard query 0xe443 AAAA DCC01		

I filtered for UDP port 5355, where it is the port for multicast. I then see in the first packets, it is like asking who is the DCC01, which is the Domain Controller. It kept on getting a response from 172.17.19.135, they are aggressively responding to every request. And a real Domain Controller would already be registered on DNS, and not use LLMNR instead. So the answer is the one always responding, **172.17.19.135**.

2. What is the hostname of the rogue machine?

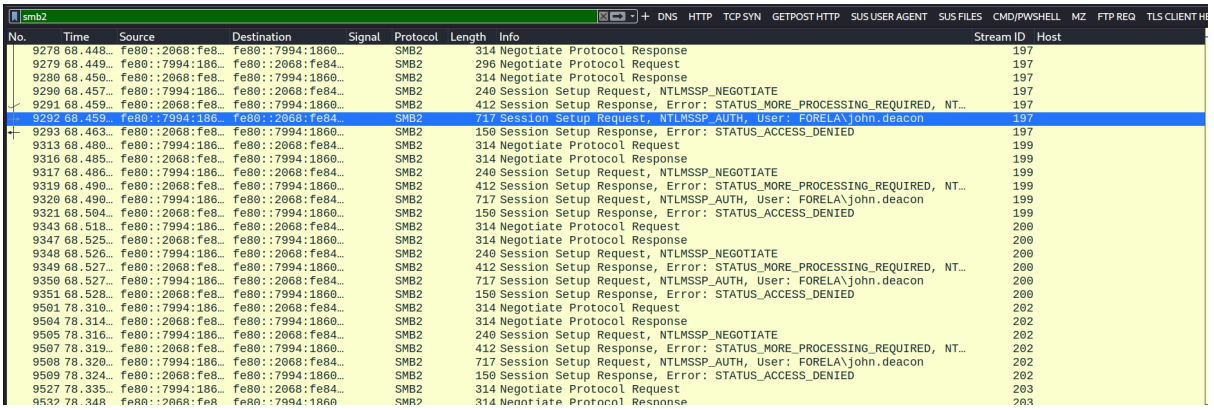


There are 3 packets for the IP of 172.17.19.135 and for DHCP to map the IP address to the host name I double clicked 1 of them.



Under the Option: Host Name dropdown we can see the host name which is **kali**

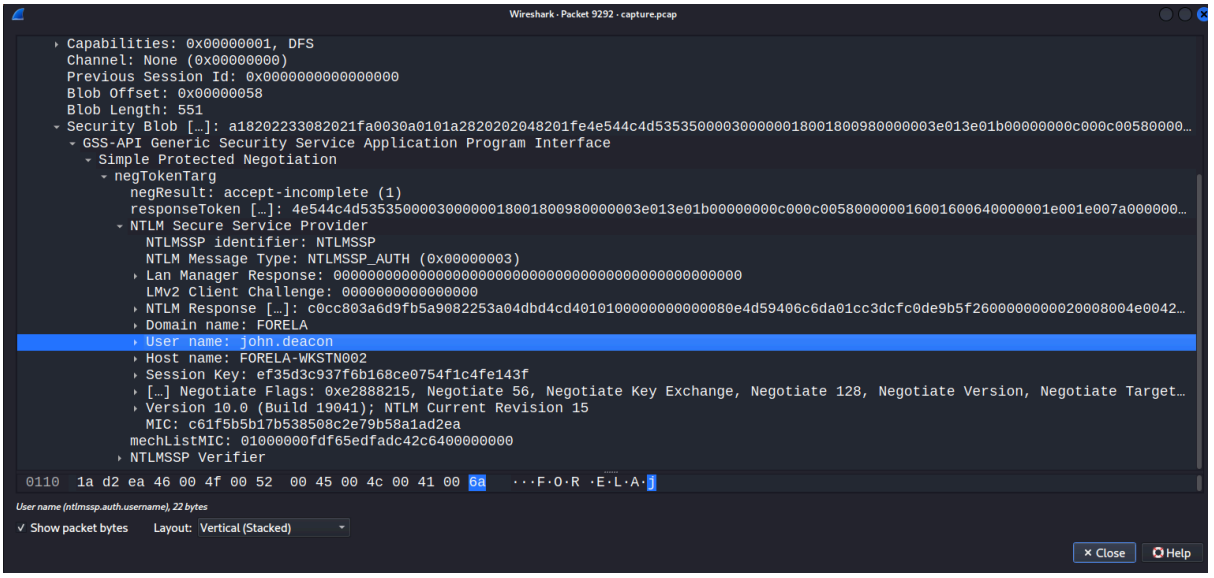
3. Now we need to confirm whether the attacker captured the user's hash and it is crackable!! What is the username whose hash was captured?



No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
9278	68.448…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	197	
9279	68.449…	fe80::7994:186…	fe80::2068:fe84…		SMB2	296	Negotiate Protocol Request	197	
9280	68.450…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	197	
9290	68.457…	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	197	
9291	68.459…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	197	
9292	68.459…	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	197	
9293	68.463…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	197	
9313	68.480…	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	199	
9316	68.485…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	199	
9317	68.486…	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	199	
9319	68.490…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	199	
9320	68.490…	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	199	
9321	68.504…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	199	
9343	68.518…	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	200	
9347	68.525…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	200	
9348	68.526…	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	200	
9349	68.527…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	200	
9350	68.527…	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	200	
9351	68.528…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	200	
9501	78.310…	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	202	
9504	78.314…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	202	
9505	78.316…	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	202	
9507	78.319…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	202	
9508	78.320…	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	202	
9509	78.324…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	202	
9527	78.335…	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	203	
9532	78.348…	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	203	

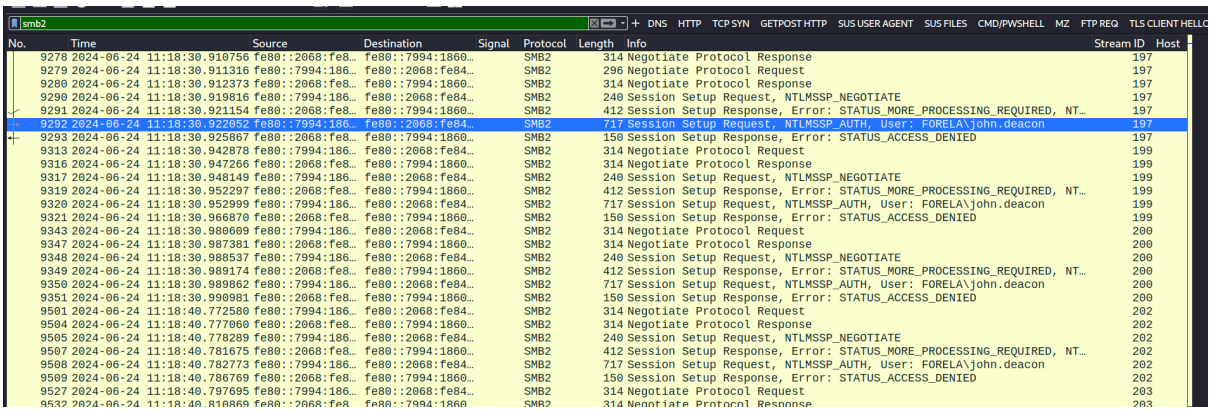
In the screenshot above, we see that I filtered for smb2, since it is the protocol used to share files within a local network. In this question SMB2 is used for authentication for NLTMSSP authentication.

In this case, when the victim wants to send something, the attacker can pretend to be legit, which is why this is a MiTM attack, and there the user gives their NLTM hash.



However, in the screenshot above, we see the username is **john.deacon**

4. In NTLM traffic we can see that the victim credentials were relayed multiple times to the attacker's machine. When were the hashes captured the First time?

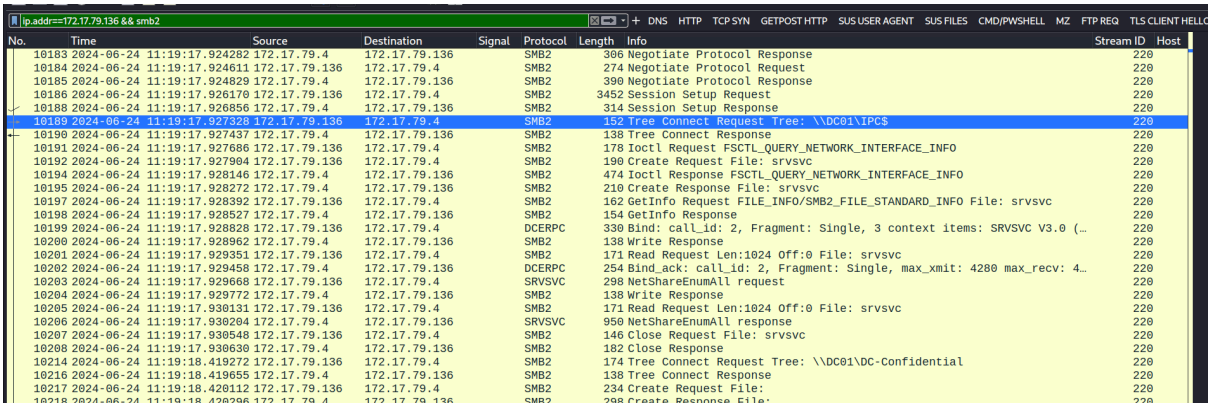


smb2

No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
9278	2024-06-24 11:18:30.910756	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	197	
9279	2024-06-24 11:18:30.911316	fe80::7994:186…	fe80::2068:fe84…		SMB2	296	Negotiate Protocol Request	197	
9280	2024-06-24 11:18:30.912373	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	197	
9290	2024-06-24 11:18:30.919816	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	197	
9291	2024-06-24 11:18:30.921154	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	197	
9292	2024-06-24 11:18:30.922052	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	197	
9293	2024-06-24 11:18:30.925867	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	197	
9313	2024-06-24 11:18:30.942878	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	199	
9316	2024-06-24 11:18:30.947266	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	199	
9317	2024-06-24 11:18:30.948149	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	199	
9319	2024-06-24 11:18:30.952297	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	199	
9320	2024-06-24 11:18:30.952999	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	199	
9321	2024-06-24 11:18:30.966870	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	199	
9343	2024-06-24 11:18:30.980609	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	200	
9347	2024-06-24 11:18:30.987381	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	200	
9348	2024-06-24 11:18:30.988537	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	200	
9349	2024-06-24 11:18:30.989174	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	200	
9350	2024-06-24 11:18:30.989862	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	200	
9351	2024-06-24 11:18:30.990981	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	200	
9501	2024-06-24 11:18:40.772580	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	202	
9504	2024-06-24 11:18:40.777060	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	202	
9505	2024-06-24 11:18:40.778289	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	202	
9507	2024-06-24 11:18:40.781675	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	202	
9508	2024-06-24 11:18:40.782773	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	202	
9509	2024-06-24 11:18:40.786769	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	202	
9527	2024-06-24 11:18:40.797695	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	203	
9532	2024-06-24 11:18:40.810869	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	203	

Still with the same screenshot from the last question and also the same packet, we can record the time when the victim gave the attacker the NLTM hash which is **2024-06-24 11:18:30**.

5. What was the typo made by the victim when navigating to the file share that caused his credentials to be leaked?



ip.addr==172.17.79.136 && smb2

No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
10183	2024-06-24 11:19:17.924282	172.17.79.4	172.17.79.136		SMB2	306	Negotiate Protocol Response	220	
10184	2024-06-24 11:19:17.924611	172.17.79.136	172.17.79.4		SMB2	274	Negotiate Protocol Request	220	
10185	2024-06-24 11:19:17.924829	172.17.79.4	172.17.79.136		SMB2	390	Negotiate Protocol Response	220	
10186	2024-06-24 11:19:17.926170	172.17.79.136	172.17.79.4		SMB2	3452	Session Setup Request	220	
10188	2024-06-24 11:19:17.926856	172.17.79.4	172.17.79.136		SMB2	314	Session Setup Response	220	
10189	2024-06-24 11:19:17.927328	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\IPC$	220	
10190	2024-06-24 11:19:17.927437	172.17.79.4	172.17.79.136		SMB2	138	Tree Connect Response	220	
10191	2024-06-24 11:19:17.927686	172.17.79.136	172.17.79.4		SMB2	178	Ioctl Request FSCTL_QUERY_NETWORK_INTERFACE_INFO	220	
10192	2024-06-24 11:19:17.927904	172.17.79.136	172.17.79.4		SMB2	190	Create Request File: srvsvc	220	
10194	2024-06-24 11:19:17.928146	172.17.79.4	172.17.79.136		SMB2	474	Ioctl Response FSCTL_QUERY_NETWORK_INTERFACE_INFO	220	
10195	2024-06-24 11:19:17.928272	172.17.79.4	172.17.79.136		SMB2	210	Create Response File: srvsvc	220	
10197	2024-06-24 11:19:17.928392	172.17.79.136	172.17.79.4		SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: srvsvc	220	
10198	2024-06-24 11:19:17.928527	172.17.79.4	172.17.79.136		SMB2	154	GetInfo Response	220	
10199	2024-06-24 11:19:17.928828	172.17.79.136	172.17.79.4		DCERPC	330	Bind: call_id: 2, Fragment: Single, 3 context items: SRVSVC V3.0 (…	220	
10200	2024-06-24 11:19:17.928962	172.17.79.4	172.17.79.136		SMB2	138	Write Response	220	
10201	2024-06-24 11:19:17.929351	172.17.79.136	172.17.79.4		SMB2	171	Read Request Len:1024 Off:0 File: srvsvc	220	
10202	2024-06-24 11:19:17.929458	172.17.79.4	172.17.79.136		DCERPC	254	Bind_ack: call_id: 2, Fragment: Single, max_xmit: 4280 max_recv: 4…	220	
10203	2024-06-24 11:19:17.929668	172.17.79.136	172.17.79.4		SRVSVC	298	NetShareEnumAll request	220	
10204	2024-06-24 11:19:17.929772	172.17.79.4	172.17.79.136		SMB2	138	Write Response	220	
10205	2024-06-24 11:19:17.930131	172.17.79.136	172.17.79.4		SMB2	171	Read Request Len:1024 Off:0 File: srvsvc	220	
10206	2024-06-24 11:19:17.930204	172.17.79.4	172.17.79.136		SRVSVC	950	NetShareEnumAll response	220	
10207	2024-06-24 11:19:17.930548	172.17.79.136	172.17.79.4		SMB2	146	Close Request File: srvsvc	220	
10208	2024-06-24 11:19:17.930630	172.17.79.4	172.17.79.136		SMB2	182	Close Response	220	
10214	2024-06-24 11:19:18.419272	172.17.79.136	172.17.79.4		SMB2	174	Tree Connect Request Tree: \\DC01\DC-Confidential	220	
10216	2024-06-24 11:19:18.419655	172.17.79.4	172.17.79.136		SMB2	138	Tree Connect Response	220	
10217	2024-06-24 11:19:18.420112	172.17.79.136	172.17.79.4		SMB2	234	Create Request File:	220	
10218	2024-06-24 11:19:18.420296	172.17.79.4	172.17.79.136		SMB2	298	Create Response File:	220	

I filtered out for the IP of the victim which is 172.17.79.136, and smb2 for the file sharing within LAN protocol and immediately I saw DC01, where it is supposed to be DCC01, so the answer is **DCC01**.

6. To get the actual credentials of the victim user we need to stitch together multiple values from the ntlm negotiation packets. What is the NTLM server challenge value?

I filtered again with smb2. I checked out the Request and Response Negotiation packets but saw nothing, therefore I saw the 1 packet before the username has been found, which is the highlighted packet in the screenshot below.

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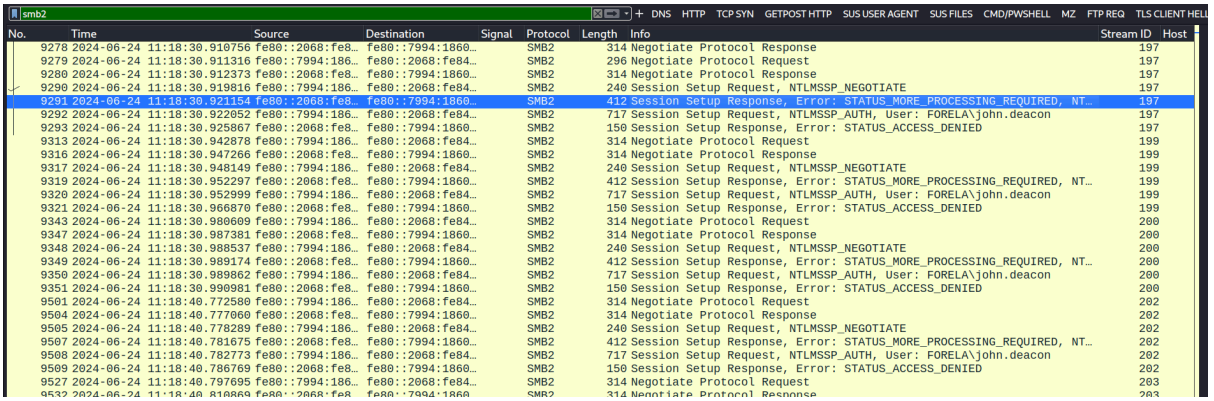
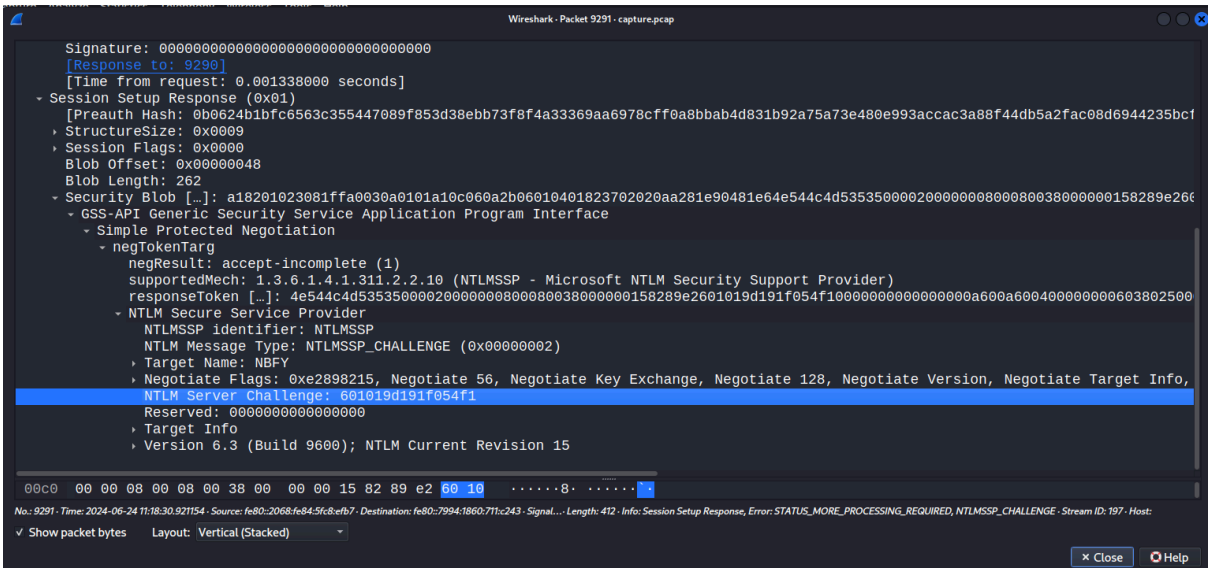


No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
9278	2024-06-24 11:18:30.910756	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	197	
9279	2024-06-24 11:18:30.911316	fe80::7994:186…	fe80::2068:fe84…		SMB2	296	Negotiate Protocol Request	197	
9280	2024-06-24 11:18:30.912373	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	197	
9290	2024-06-24 11:18:30.919816	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	197	
9291	2024-06-24 11:18:30.921154	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	197	
9292	2024-06-24 11:18:30.922052	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	197	
9293	2024-06-24 11:18:30.925867	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	197	
9313	2024-06-24 11:18:30.942878	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	199	
9316	2024-06-24 11:18:30.947266	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	199	
9317	2024-06-24 11:18:30.948149	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	199	
9319	2024-06-24 11:18:30.952297	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	199	
9320	2024-06-24 11:18:30.952999	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	199	
9321	2024-06-24 11:18:30.966870	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	199	
9343	2024-06-24 11:18:30.980609	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	200	
9347	2024-06-24 11:18:30.987381	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	200	
9348	2024-06-24 11:18:30.988537	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	200	
9349	2024-06-24 11:18:30.989174	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	200	
9350	2024-06-24 11:18:30.989862	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	200	
9351	2024-06-24 11:18:30.990981	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	200	
9501	2024-06-24 11:18:40.772580	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	202	
9504	2024-06-24 11:18:40.777060	fe80::2068:fe8…	fe80::7994:1860…		SMB2	314	Negotiate Protocol Response	202	
9505	2024-06-24 11:18:40.778289	fe80::7994:186…	fe80::2068:fe84…		SMB2	240	Session Setup Request, NTLMSSP_NEGOTIATE	202	
9507	2024-06-24 11:18:40.781675	fe80::2068:fe8…	fe80::7994:1860…		SMB2	412	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	202	
9508	2024-06-24 11:18:40.782773	fe80::7994:186…	fe80::2068:fe84…		SMB2	717	Session Setup Request, NTLMSSP_AUTH, User: FORELA\john.deacon	202	
9509	2024-06-24 11:18:40.786769	fe80::2068:fe8…	fe80::7994:1860…		SMB2	150	Session Setup Response, Error: STATUS_ACCESS_DENIED	202	
9527	2024-06-24 11:18:40.797695	fe80::7994:186…	fe80::2068:fe84…		SMB2	314	Negotiate Protocol Request	203	

Then I double clicked on the packet:

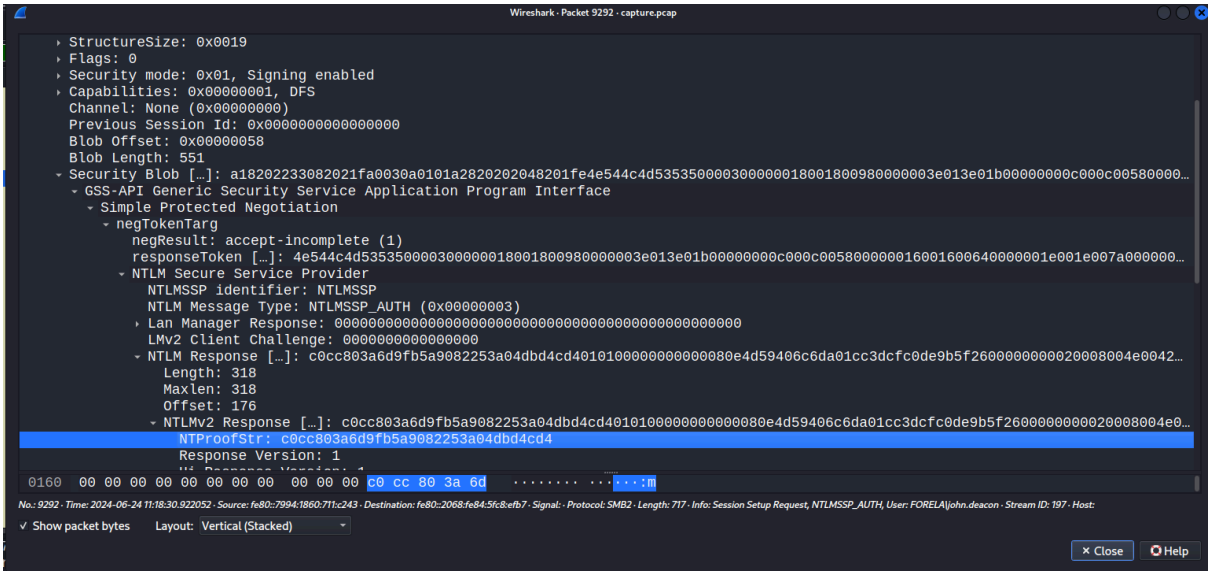


```
Signature: 00000000000000000000000000000000
[Response to: 9290]
[Time from request: 0.001338000 seconds]
Session Setup Response (0x01)
[Preauth Hash: 0b0624b1bfc6563c355447089f853d38ebb73f8f4a33369aa6978cff0a8bbab4d831b92a75a73e480e993accac3a88f44db5a2fac08d6944235bcf
StructureSize: 0x0009
Session Flags: 0x0000
Blob Offset: 0x00000048
Blob Length: 262
Security Blob […]: a18201023081ffa0030a0101a10c060a2b06010401823702020aa281e90481e64e544c4d5353500002000000080008003800000158289e266
GSS-API Generic Security Service Application Program Interface
Simple Protected Negotiation
negTokenTarg
negResult: accept-incomplete (1)
supportedMech: 1.3.6.1.4.1.311.2.2.10 (NTLMSSP - Microsoft NTLM Security Support Provider)
responseToken […]: 4e544c4d5353500002000000080008003800000158289e2601019d191f054f1000000000000000a600a60040000000603802500
NTLM Secure Service Provider
NTLMSSP identifier: NTLMSSP
NTLM Message Type: NTLMSSP_CHALLENGE (0x00000002)
Target Name: NBFY
Negotiate Flags: 0xe2898215, Negotiate 56, Negotiate Key Exchange, Negotiate 128, Negotiate Version, Negotiate Target Info,
NTLM Server Challenge: 601019d191f054f1
Reserved: 0000000000000000
Target Info
Version 6.3 (Build 9600); NTLM Current Revision 15

00c0  00 00 08 00 08 00 38 00  00 00 15 82 89 e2 60 10   ·····8· ····· `·
```

There I found the NLTM server challenge which is **601019d191f054f1**.

7. Now doing something similar find the NTProofStr value.

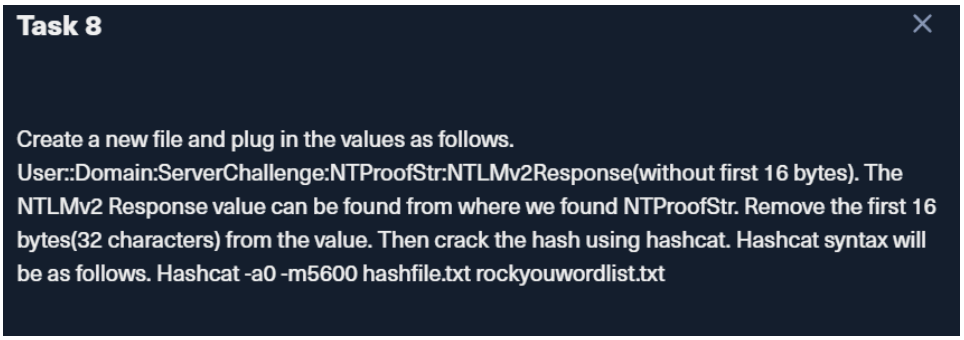


```
StructureSize: 0x0019
Flags: 0
Security mode: 0x01, Signing enabled
Capabilities: 0x00000001, DFS
Channel: None (0x00000000)
Previous Session Id: 0x0000000000000000
Blob Offset: 0x00000058
Blob Length: 551
Security Blob […]: a18202233082021fa0030a0101a2820202048201fe4e544c4d5353500003000000180018009800000003e013e01b00000000c000c00580000…
GSS-API Generic Security Service Application Program Interface
Simple Protected Negotiation
negTokenTarg
negResult: accept-incomplete (1)
responseToken […]: 4e544c4d5353500003000000180018009800000003e013e01b00000000c000c00580000001600160064000000 1e001e007a000000…
NTLM Secure Service Provider
NTLMSSP identifier: NTLMSSP
NTLM Message Type: NTLMSSP_AUTH (0x00000003)
Lan Manager Response: 00000000000000000000000000000000000000000000000000
LMv2 Client Challenge: 0000000000000000
NTLM Response […]: c0cc803a6d9fb5a9082253a04dbd4cd40101000000000000080e4d59406c6da01cc3dcfc0de9b5f2600000000020008004e00 42…
Length: 318
Maxlen: 318
Offset: 176
NTLMv2 Response […]: c0cc803a6d9fb5a9082253a04dbd4cd40101000000000000080e4d59406c6da01cc3dcfc0de9b5f2600000000020008004e0…
NTProofStr: c0cc803a6d9fb5a9082253a04dbd4cd4
Response Version: 1

0160  00 00 00 00 00 00 00 00  00 00 00 c0 cc 80 3a 6d   ····· ····· ····:m
```

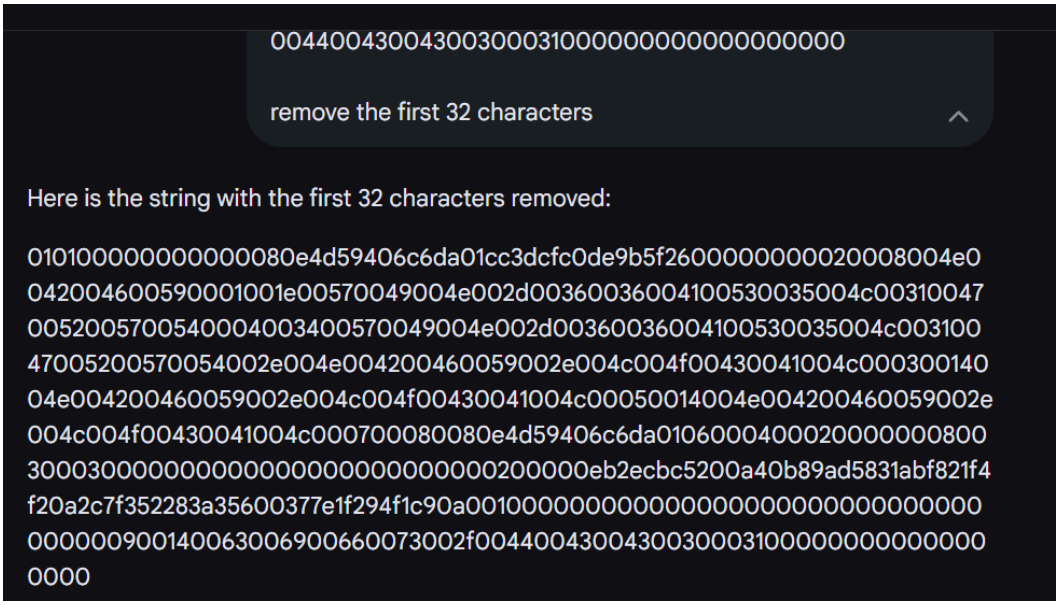
Going back to packet number 9292, where we found the host name of the victim, we see a the NTProofStr attribute, and for this question, the answer is **c0cc803a6d9fb5a9082253a04dbd4cd4**.

8. To test the password complexity, try recovering the password from the information found from packet capture. This is a crucial step as this way we can find whether the attacker was able to crack this and how quickly.



For this question, I opened the hint and it asked me to get the NLTMv2 response value which I got it here like shown in the screenshot below

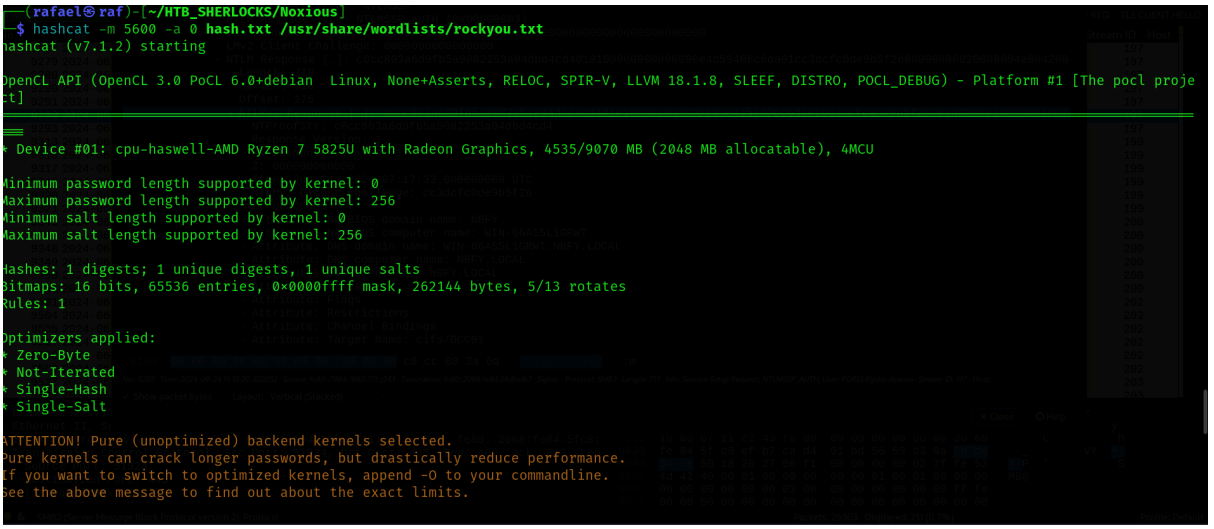
And I asked Google to reduce the first 32 characters of the string:



Combining it with the host name, and the domain name here is what I put for hash.txt.

John.deacon::FORELA:601019d191f054f1:c0cc803a6d9fb5a9082253a04dbd4cd4:010100000000000080e4d59406c6da01cc3dcfc0de9b5f2600000000020008004e004200460059000 1001e00570049004e002d003600360041

00530035004c003100470052005700540004003400570049004e002d003
6003600410053003 5004c003100470052005700540 02e004e0042004600
59002e004c004f00430041004c00030014004e004200460059002e004c00
4f00430041004c00050014004e004200460059002e004c004f0043004100
4c000700080080e4d59406c6da0106000400020000000800300030000000
00000000000000000000200000eb2ecbc5200a40b89ad5831abf821f4f20a2
c7f352283a35600377e1f294f1c90a00100000000000000000000000000000
00000000900140063006900660073002f00440043004300300031000000
000000000000



Then I used hashcat to know the string behind the hash.

And here we can see the password which is

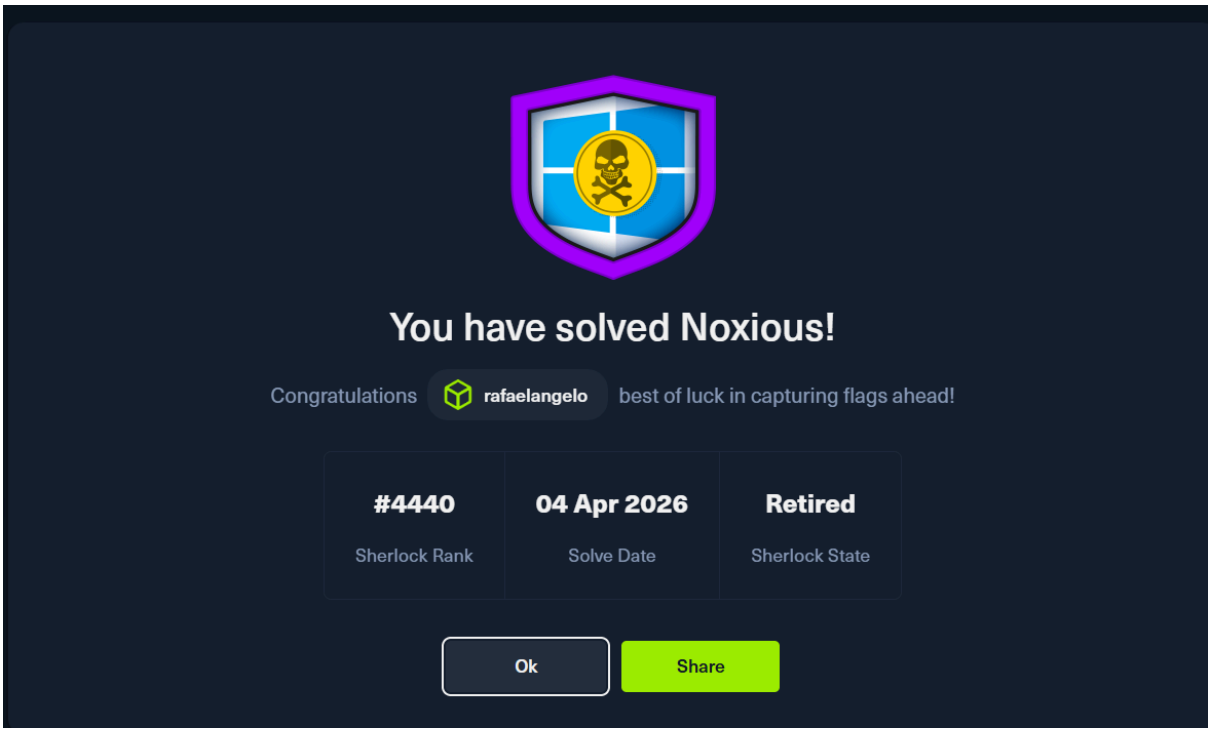
9. Just to get more context surrounding the incident, what is the actual file share that the victim was trying to navigate to?

ip.addr == 172.17.79.136 && smb2

No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID
10184	2024-06-24 11:19:17.924611	172.17.79.136	172.17.79.4		SMB2	274	Negotiate Protocol Request	22
10186	2024-06-24 11:19:17.926170	172.17.79.136	172.17.79.4		SMB2	3452	Session Setup Request	22
10189	2024-06-24 11:19:17.927328	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\IPC$	22
10191	2024-06-24 11:19:17.927686	172.17.79.136	172.17.79.4		SMB2	178	Ioctl Request FSCTL_QUERY_NETWORK_INTERFACE_INFO	22
10192	2024-06-24 11:19:17.927904	172.17.79.136	172.17.79.4		SMB2	190	Create Request File: srvsvc	22
10197	2024-06-24 11:19:17.928392	172.17.79.136	172.17.79.4		SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: srvsvc	22
10199	2024-06-24 11:19:17.928828	172.17.79.136	172.17.79.4		DCERPC	330	Bind: call_id: 2, Fragment: Single, 3 context items: SRVSVC V3.0 (…	22
10201	2024-06-24 11:19:17.929351	172.17.79.136	172.17.79.4		SMB2	171	Read Request Len:1024 Off:0 File: srvsvc	22
10203	2024-06-24 11:19:17.929668	172.17.79.136	172.17.79.4		SRVSVC	298	NetShareEnumAll request	22
10205	2024-06-24 11:19:17.930131	172.17.79.136	172.17.79.4		SMB2	171	Read Request Len:1024 Off:0 File: srvsvc	22
10207	2024-06-24 11:19:17.930548	172.17.79.136	172.17.79.4		SMB2	146	Close Request File: srvsvc	22
10214	2024-06-24 11:19:18.419272	172.17.79.136	172.17.79.4		SMB2	174	Tree Connect Request Tree: \\DC01\DC-Confidential	22
10217	2024-06-24 11:19:18.420112	172.17.79.136	172.17.79.4		SMB2	234	Create Request File:	22
10219	2024-06-24 11:19:18.420617	172.17.79.136	172.17.79.4		SMB2	146	Close Request File:	22
10221	2024-06-24 11:19:18.421353	172.17.79.136	172.17.79.4		SMB2	190	Create Request File: srvsvc	22
10223	2024-06-24 11:19:18.421999	172.17.79.136	172.17.79.4		SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: srvsvc	22
10225	2024-06-24 11:19:18.422625	172.17.79.136	172.17.79.4		DCERPC	330	Bind: call_id: 2, Fragment: Single, 3 context items: SRVSVC V3.0 (…	22
10227	2024-06-24 11:19:18.422893	172.17.79.136	172.17.79.4		SMB2	171	Read Request Len:1024 Off:0 File: srvsvc	22
10229	2024-06-24 11:19:18.423522	172.17.79.136	172.17.79.4		SRVSVC	302	NetShareGetInfo request	22
10231	2024-06-24 11:19:18.423806	172.17.79.136	172.17.79.4		SMB2	171	Read Request Len:1024 Off:0 File: srvsvc	22
10233	2024-06-24 11:19:18.424343	172.17.79.136	172.17.79.4		SMB2	146	Close Request File: srvsvc	22
10235	2024-06-24 11:19:18.447545	172.17.79.136	172.17.79.4		SMB2	366	Create Request File:	22
10237	2024-06-24 11:19:18.448383	172.17.79.136	172.17.79.4		SMB2	382	Create Request File: desktop.ini	22
10239	2024-06-24 11:19:18.448712	172.17.79.136	172.17.79.4		SMB2	306	Create Request File: ;Notify Request	22
10240	2024-06-24 11:19:18.449102	172.17.79.136	172.17.79.4		SMB2	412	Create Request File: ;Find Request SMB2_FIND_ID_BOTH_DIRECTORY_INF…	22
10243	2024-06-24 11:19:18.449613	172.17.79.136	172.17.79.4		SMB2	306	Create Request File: ;Notify Request	22

Since it has something to do with the victim I filtered for their IP address (172.17.79.135), and smb2 since it has to do with file sharing within LAN.

In the packets above we see as well "\\DC01\IPC$", and while I thought that was the answer, I was wrong, since IPC$ is a default system share used for communication, which can be used for authentication and session setup. While "**\\DC01\DC-Confidential**" is the actual file the victim is trying to navigate to.



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